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ADHD DETECTION WITH EEG SIGNALS

Submitted for the course **Information Theory and Feature Engineering**

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Declaration

I declare that I have written this report by myself and have only used the sources and aids mentioned, and that I have marked direct and indirect citations as such. This report has not been submitted prior for any other examination.

I agree that the results of this study work / report may be used free of charge for research and lecturing purposes.

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List of Abbreviations

AC	Alternating Current
ADHD	Attention-Deficit/Hyperactivity Disorder
DC	Direct Current
EEG	Electroencephalography

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental condition that affects individuals across the lifespan. According to Furman [1], ADHD is not a single disease entity but rather a constellation of symptoms representing a final common behavioral pathway for a range of emotional, psychological, and learning difficulties.

Although the behavioral manifestations of ADHD are well-documented, its neurophysiological underpinnings remain an active area of research. Electroencephalography (EEG) has been widely used to study ADHD, as it provides a non-invasive measure of brain electrical activity. Previous studies have identified characteristic EEG patterns in individuals with ADHD, particularly abnormalities in the frontal and central regions, often reflected as altered spectral power or atypical signal complexity.

This study aims to investigate differences in EEG activity between healthy individuals and those diagnosed with ADHD, using the dataset provided by Sadeghi Bajestani *et al.* [2]. Specifically, the project focuses on (1) comparing the statistical and informational properties of EEG signals across both groups, (2) engineering features inspired by information theory—such as entropy and mutual information—to quantify signal complexity and connectivity, and (3) building a predictive model capable of distinguishing ADHD from healthy controls based on these features. This work bridges the domains of signal processing, feature engineering, and information theory to improve understanding and potential classification of ADHD-related brain activity.

2 Data Description

The dataset [2] comprises EEG recordings from a total of 79 adult participants, including 42 healthy controls and 37 individuals diagnosed with ADHD (age range: 20-68 years; male/female ratio: 56/23). EEG signals were recorded under four experimental conditions: (1) resting state with eyes open, (2) resting state with eyes closed, (3) a cognitive challenge task, and (4) an auditory task involving listening to an “omni harmonic” sound stimulus. Recordings were obtained from five scalp locations—O1, F3, F4, Cz, and Fz—at a sampling rate of 256 Hz. These regions encompass occipital and frontal areas known to play key roles in attentional control and executive function.

File organization and structure. The data are provided as four MATLAB `.mat` files, corresponding to the experimental groups:

- `FC.mat` — female control group
- `MC.mat` — male control group
- `FADHD.mat` — female ADHD group
- `MADHD.mat` — male ADHD group

Each file contains a 1×11 cell array, where each cell represents one experimental task or condition. Within each cell, the data are stored as a three-dimensional matrix with dimensions `[subjects×samples×channels]`. For instance, a typical entry of size $13 \times 7680 \times 2$ indicates 13

participants, 7680 time samples (corresponding to 30 seconds of EEG data at 256 Hz), and two recorded channels.

The specific configuration of each cell (i.e., channel pair and duration) is summarized as follows:

Cell	Condition	Channels	Duration (s)
1	Eyes open baseline	Cz, F4	30
2	Eyes closed	Cz, F4	20
3	Eyes open	Cz, F4	20
4	Cognitive challenge	Cz, F4	45
5	Pre-Omni harmonic baseline	Cz, F4	15
6	Omni harmonic assessment	Cz, F4	30
7	Eyes open baseline	O1, F4	30
8	Eyes closed	O1, F4	30
9	Eyes open	O1, F4	30
10	Eyes closed	F3, F4	45
11	Eyes closed	Fz, F4	45

Table 1: Summary of EEG tasks, channel pairs, and recording durations.

One corrupted signal (subject 7 of the female ADHD group) was identified and excluded from further analysis.

This hierarchical data organization allows for flexible analysis across multiple dimensions: by gender, diagnosis, condition, and channel pair. For subsequent preprocessing, each task will be reshaped into individual subject-channel recordings, ensuring consistent sampling durations across conditions.

3 Data Visualization

3.1 Data Loading and Structure

To analyze EEG signals from the ADHD dataset [2], the recordings were imported from MATLAB `.mat` files using the `scipy.io.loadmat` library in Python. Four files were available, corresponding to each experimental group: `FC.mat` (female control), `MC.mat` (male control), `FADHD.mat` (female ADHD), and `MADHD.mat` (male ADHD). Each file contained a 1×11 cell array, where each cell represented a specific experimental task and stored EEG data as a three-dimensional matrix with dimensions [subjects x samples x channels].

A custom loading function (`load_group`) was implemented to iterate through each cell, extract signal arrays, and store them in a structured pandas `DataFrame`. Each row in the resulting dataset corresponds to a single subject-task pair and includes the following metadata: participant group (Control/ADHD), gender, task number, subject identifier, and a two-channel EEG signal array. This organization facilitates subsequent preprocessing, visualization, and feature extraction. The corrupted EEG file corresponding to subject 7 of the female ADHD group was identified and excluded from analysis.

3.2 Signal Visualization by Group and Task

To inspect data quality and explore inter-subject variability, raw EEG signals were visualized for selected participants using `matplotlib`. Figure 1 presents the recordings from the first five female control subjects during Task 1 (eyes-open baseline). Each subplot represents one subject, showing the two recorded channels.

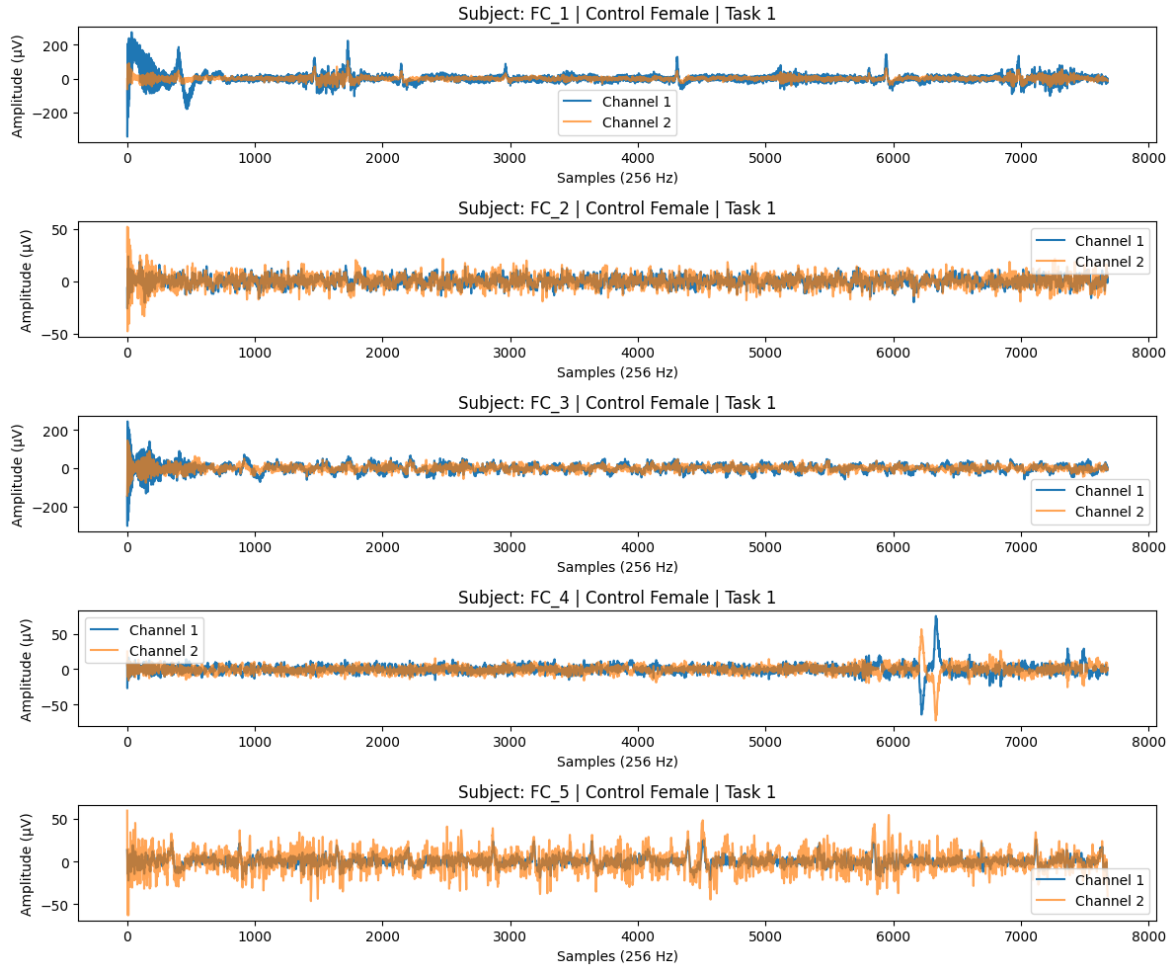


Figure 1: EEG signals from five female control participants during Task 1 (eyes-open baseline). Each subplot represents two simultaneously recorded channels.

As illustrated in Figure 1, the EEG traces display characteristic oscillatory patterns within an amplitude range of approximately $\pm 200 \mu\text{V}$. Variations across participants are evident, with some showing strong low-frequency drifts or transient spikes, likely associated with ocular or muscular artifacts. Despite these variations, both channels exhibit similar temporal trends, suggesting functional coupling between the recorded electrode sites. These observations emphasize the need for further preprocessing steps, such as band-pass filtering (0.5 — 45 Hz), baseline correction, and normalization, to ensure consistent signal quality across participants.

3.3 Comparison Between ADHD and Control Groups

A complementary visualization was conducted to qualitatively compare EEG signals between ADHD and control subjects under the same task conditions. Figure 2 shows five pairs of male participants (Control vs. ADHD) performing Task 1, for Channel 0.

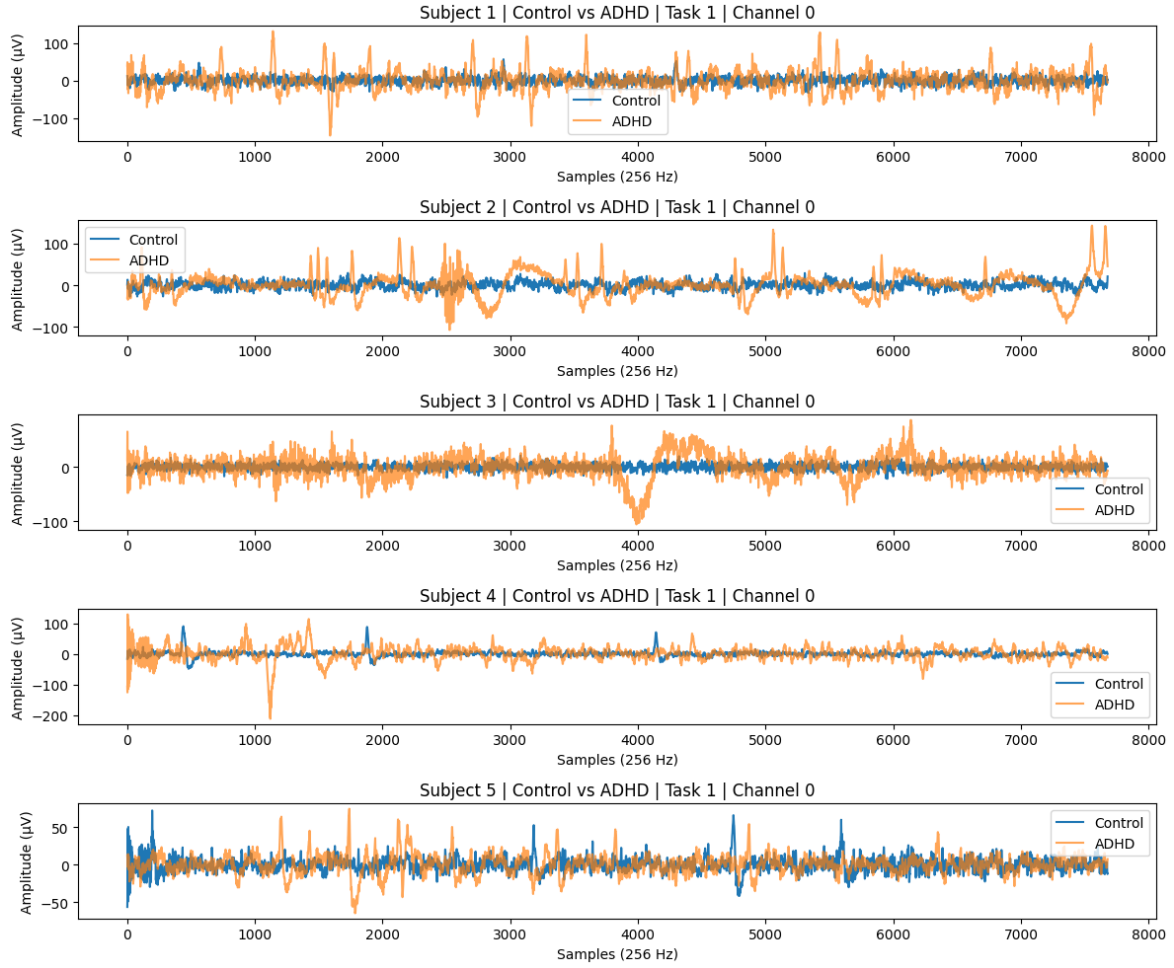


Figure 2: Comparison of EEG signals between control and ADHD male participants during Task 1 for Channel 0. Each subplot shows a pair of subjects (Control vs. ADHD).

In general, ADHD recordings exhibit greater amplitude fluctuations and reduced rhythmic stability compared to control subjects, who display smoother oscillatory activity. This increased irregularity suggests higher signal entropy and reduced neural synchronization in ADHD participants. Such qualitative patterns are consistent with prior research reporting atypical cortical activity and impaired oscillatory regulation in ADHD populations [3]. These visual differences motivate the extraction of quantitative features — particularly information-theoretic measures such as entropy and mutual information — to characterize signal complexity and connectivity.

3.4 Observations

The exploratory visualization phase revealed several important insights:

- EEG signals contain observable low-frequency drifts and high-amplitude artifacts, supporting the need for filtering and normalization.
- There is substantial inter-subject variability in signal amplitude and frequency content, even within the same condition.
- ADHD participants tend to exhibit higher signal variability and irregularity, indicating potential discriminative features for subsequent classification.

These findings informed the design of the preprocessing and feature extraction pipeline, where statistical, spectral, and information-theoretic features were computed to enable quantitative group comparison and machine learning-based classification.

4 Preprocessing

EEG data are highly sensitive to non-neural artifacts arising from muscle activity, eye blinks, and small head or body movements. Because the dataset used in this study was recorded non-invasively from scalp electrodes, such artifacts are inevitable and must be carefully mitigated to preserve the validity of the neural information [4]. The objective of the preprocessing stage was therefore to reduce non-cerebral noise to isolate the neural dynamics relevant to ADHD.

Each recording was first detrended by removing its mean to eliminate slow DC components. A notch filter at 50 Hz was used to attenuate power-line interference and fourth-order Butterworth band-pass filter between 0.5 Hz and 45 Hz was then applied to suppress slow drifts and high-frequency muscular or environmental noise while preserving the main EEG frequency bands (delta, theta, alpha, beta, and low-gamma). In addition, a notch filter at 50 Hz was used to attenuate power-line interference.

4.1 Detrending

EEG recordings were detrended by subtracting the mean amplitude of each channel to remove Direct Current (DC) offsets and slow baseline drifts [5]. EEG recordings often contain a small DC offset — a constant voltage bias caused by electrode-skin impedance and amplifier drift changes. This offset shifts the entire signal above or below zero microvolts without carrying any neural information. This step prevents artificial low-frequency power and ensures that subsequent analyses reflect genuine neural oscillations. Although the visual appearance of the waveforms remains largely unchanged, detrending is essential to prevent bias in later spectral and entropy-based feature extraction.

To verify the effectiveness of the detrending procedure, the mean amplitude of each EEG channel was computed before and after baseline correction. As shown below in table 2, the mean values were reduced to approximately zero following detrending, confirming successful removal of the DC offset while preserving the oscillatory structure of the signal.

	Channel 1	Channel 2
Mean before detrending (μV)	1.5269	0.0936
Mean after detrending (μV)	-1.74×10^{-15}	1.18×10^{-16}

Table 2: Verification of mean amplitude before and after detrending.

The results confirm that the detrending operation correctly centered the EEG signals around zero microvolts, thereby eliminating slow baseline drifts without altering the underlying temporal dynamics. This ensures that subsequent frequency and entropy analyses are not biased by non-neural voltage offsets.

4.2 Notch Filtering

During EEG acquisition, recordings are often contaminated by electrical noise originating from alternating current (AC) power lines.

Through capacitive coupling, this strong electrical signal “leaks” into the EEG recording. Since brain waves are measured in microvolts (μV) and power lines carry huge voltage, this “hum” can completely drown out brain signals, especially in the Gamma band (30-50 Hz).

To eliminate main electricity noise, an IIR Notch Filter was applied at 50 Hz. The quality factor was set to $Q = 30$ to minimize the distortion of surrounding frequencies while effectively suppressing the power line hum.

4.3 Band-Pass Filtering

A crucial step in EEG preprocessing is the removal of frequency components that do not reflect neural activity. Raw EEG recordings are highly susceptible to slow drifts caused by skin potentials and electrode impedance fluctuations, as well as high-frequency contamination originating from muscle activity (EMG), electrical noise, and environmental interference. If not addressed, these artifacts can dominate the signal and bias subsequent feature extraction, particularly when computing spectral or information-theoretic measures.

To mitigate these issues, each EEG segment was band-pass filtered between 1 Hz and 50 Hz using a 2nd order Butterworth filter. Frequencies below 1 Hz primarily capture non-neural low-frequency drift, while components above 50 Hz are largely associated with muscular activity and electrical noise.

4.4 Preprocessing Comparison

Figure 3 illustrates a representative example comparing the data before and after the preprocessing, in this example is the female group control.

The left panel displays the raw EEG signal, exhibiting baseline drift and high-frequency noise. The right panel shows the signal after Detrending, Notch Filtering (50Hz), and Bandpass Filtering (1-40Hz). After preprocessing, the signal is centered around zero, and high-frequency fluctuations associated with power-line noise are visibly attenuated. Importantly, the dominant oscillatory patterns remain intact, indicating that preprocessing reduced noise without distorting meaningful neural activity.

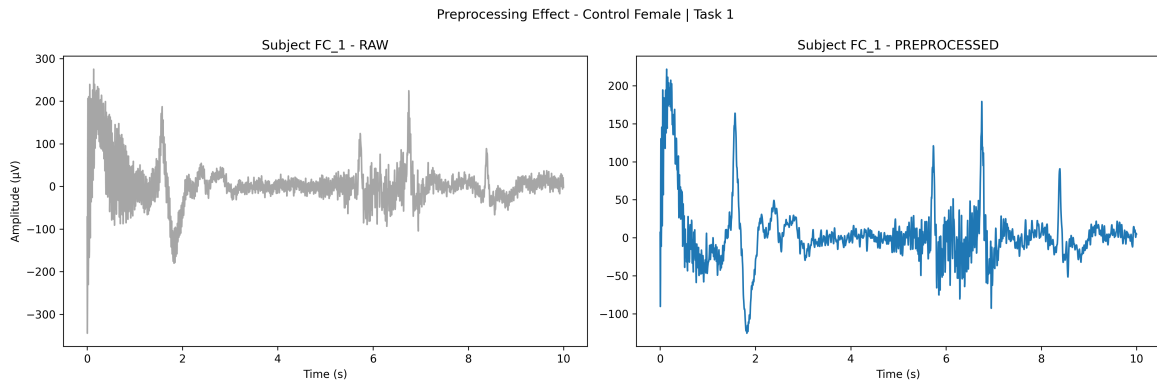


Figure 3: Preprocessing Comparisson

4.5 Quantitative Comparison

To quantitatively evaluate the impact of preprocessing, several signal quality metrics were computed before and after filtering. These included signal variance, 50 Hz line noise power, signal-to-noise ratio (SNR), and Shannon entropy:

1. **Total Variance Reduction:** Since ocular and muscle artifacts typically manifest as high-amplitude transients significantly larger than neural potentials, a reduction in signal variance serves as a global proxy for artifact suppression.
2. **50 Hz Line Noise Power:** We computed the integrated power in the 49–51 Hz band to specifically validate the efficacy of the IIR Notch filter in eliminating mains interference.
3. **Signal-to-Noise Ratio (SNR):** SNR was calculated as the ratio of power in the physiological EEG band (1–45 Hz) to the power in artifactual bands (0.1–1 Hz drift and > 55 Hz muscle noise). An increase in SNR indicates that the filtering preserved neural information while selectively attenuating noise.
4. **Shannon Entropy:** Entropy measures the complexity or randomness of a signal. Since electromyogenic (muscle) noise often resembles random processes, a shift in entropy post-processing reflects the recovery of structured, rhythmic neural oscillations (e.g., Alpha waves).

Metric	Raw Signal (Mean)	Preprocessed (Mean)	Change / Improvement
Total Variance (μV^2)	242.6	191.4	21.1% Reduction
50 Hz Power (Line Noise)	20.62	0.29	98.6% Suppression
High-Freq Noise (> 55Hz)	1.67	0.12	92.8% Suppression
Signal-to-Noise Ratio (SNR)	13.65	21.82	59.8% Improvement
Shannon Entropy	2.75	2.80	Stable (+1.8%)

Table 3: Quantitative Assessment of Preprocessing Efficacy (Averaged across N=13 subjects)

Table ?? presents the quantitative impact of the preprocessing pipeline. The most significant improvement was observed in the removal of power line interference, with a 98.6% reduction in 50 Hz power. The 92.8% reduction in high-frequency noise suggests that electromyogenic (muscle) artifacts, which predominantly affect the gamma band, were effectively suppressed

Notably, the total signal variance decreased by only 21.1%, while the specific noise components were nearly eliminated. This indicates that the preprocessing pipeline successfully preserved the majority of the low-frequency physiological brain activity (Delta, Theta, Alpha) while surgically removing environmental artifacts. Consequently, the Signal-to-Noise Ratio (SNR) improved from 13.65 to 21.82, providing a cleaner input for the subsequent feature extraction stages."

References

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AI Prompt Used

1. Give me a good structure for the a report in this.
2. What are the best preprocessing methods I can use for this dataset?
3. Are PCA, WT or EMD good methods for preprocessing my data?